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Optimal networks for excitonic energy transport

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Abstract
We investigate coherent and incoherent excitation transfer in a random network with dipole–dipole interactions as a model system describing energy transport, e.g., in photosynthetic light-harvesting complexes or gases of cold Rydberg atoms. For this purpose, we introduce and compare two different measures (the maximum output probability and the average transfer time) for the efficiency of transport from the input to the output site. We especially focus on optimal configurations which maximize the transfer efficiency and the impact of dephasing noise on the transport dynamics. For most configurations of the random network, the transfer efficiency increases when adding noise, giving rise to essentially classical transport. These noise-assisted configurations are, however, systematically less efficient than the optimal configurations. The latter reach their highest efficiency for purely coherent dynamics, i.e. in the absence of noise.

1. Introduction

A current and actively discussed issue is the mechanisms that drive efficient transport across photosynthetic complexes. Recent experiments (Engel *et al* 2007) detected long-lived quantum coherence in the Fenna–Matthews–Olson (FMO) complex of green sulfur bacteria. This discovery refutes the long-accepted belief that, in spite of the quantum nature of the very fabric of life (atoms and molecules), quantum effects would nevertheless be irrelevant for most biological processes, simply because they operate at room temperature and involve vastly many degrees of freedom.

The FMO complex carries absorbed solar energy to a reaction centre, where it is transformed to chemical energy that fuels the bacterium. With only seven (or eight (Olbrich *et al* 2010, Schmidt am Busch *et al* 2011)) molecules participating in the excitation transport, the FMO complex is the simplest and smallest light-harvesting structure in nature. It features remarkably high light-to-charge conversion rates of more than 95% (Cheng and Fleming 2009). Since classical theory is unable to explain this high efficiency, we are tempted to suspect that the recently detected quantum coherence is behind the efficient dynamics.

Excitations in the FMO complex behave, in many respects, similar to single excitations in gases of cold Rydberg atoms (Gallagher 2005, Pohl *et al* 2008, Weatherill *et al* 2008, Younge *et al* 2009): in both cases, transfer of an excitation between two molecules of the light harvesting complex or, respectively, between two Rydberg atoms occurs by near-resonant dipole–dipole interaction. Also in the latter case, quantum coherence is expected to affect the transport properties (Anderson *et al* 1998, Mourachko *et al* 1998, Akulin *et al* 1999, Westermann *et al* 2006, Carroll *et al* 2006, Ates *et al* 2007, Mülken *et al* 2007), since for Rydberg atoms, decoherence due to spontaneous emission is very slow compared to the timescale of transport induced by the dipole–dipole interaction.

Generally, one may then ask whether quantum interference leads rather to enhancement or suppression of transport as compared to the classical incoherent dynamics. In many situations, one finds a destructive impact, e.g. disordered lattices with nearest-neighbour coupling in one or two dimensions (to give just one example) exhibit exponentially localized eigenstates—a phenomenon known as Anderson localization (Anderson 1958). In this case, the excitation forever remains in the vicinity of the initial site, whereas, according to the classical prediction, it would diffuse to arbitrarily distant sites in the limit of long times. Also for smaller systems—i.e. smaller than the localization length defined for an infinite system—inhibition of transport

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due to destructive interference is frequently encountered, for example, one of the best currently available model Hamiltonians for the FMO complex (Adolphs and Renger 2006) leads to rather inefficient coherent transport (Scholak *et al* 2011, 2010). Therefore, it has been argued (Rebentrost *et al* 2009, Plenio and Huelga 2008) that additional classical noise is needed in order to increase the transport efficiency ('environment-assisted transport' (Plenio and Huelga 2008, Caruso *et al* 2009, Rebentrost *et al* 2009, Cao and Silbey 2009, Wu *et al* 2010, Chin *et al* 2010)). The fact that destructive interference is observed in many examples, however, does not rule out the possibility that, in other cases, the opposite effect—enhancement of transport by constructive interference—could also occur.

In this paper, we will examine this question by means of a statistical approach that comprises many different configurations—and correspondingly different transport properties—of a quantum network with dipole–dipole-like interactions. Indeed, the precise configuration of a system under study is often not exactly known: in a Rydberg gas, for example, the atoms are located at random positions, and also the FMO model Hamiltonian mentioned above exhibits experimental uncertainties. We therefore define a statistical ensemble of Hamiltonians in section 2 and introduce quantitative measures in order to judge their transport efficiencies in section 3. In section 4, we will focus on optimal configurations which yield the highest transport efficiencies, whereas statistical properties of the whole ensemble will be discussed in section 5. Finally, in section 6 we analyse the impact of noise, in order to compare the scenarios of environment-assisted and interference-assisted transport, respectively.

2. Random quantum network

Although most systems in which energy transport occurs, such as solar cells or light-harvesting complexes, are typically composed of a large number of microscopic constituents, they can often be effectively described in terms of relatively few relevant degrees of freedom: for example, in a light harvesting complex, each molecule is reduced to its ground state and one excited state, so that the entire collection of complex macromolecules is described in terms of a few coupled two-level systems; similarly, considering transport of excitations in a gas of cold Rydberg atoms, each of the atoms can often be reduced to those two atomic levels whose transition frequency is closest to a given frequency. A typical model Hamiltonian is then of the form

$$H = \sum_{i=1}^{N} E_i\, \sigma_+^{(i)}\sigma_-^{(i)} + \sum_{i\neq j}^{N} V_{i,j}\, \sigma_+^{(i)}\sigma_-^{(j)}, \tag{1}$$

where N denotes the number of sites ($N = 7$ in case of the photosynthetic FMO complex), and $\sigma_+^{(i)}\sigma_-^{(i)}$ the projector onto the excited state of site i with excitation energy E_i. The operator $\sigma_+^{(i)}\sigma_-^{(j)}$ describes the transfer of an excitation from site j to site i with corresponding coupling strength $V_{i,j} = V_{j,i}^*$.

According to equation (1), the total number of excitations $\sum_i \sigma_+^{(i)}\sigma_-^{(i)}$ is conserved, which is characteristic for near-resonant exchange processes. Starting initially with a single excitation, we can, therefore, reduce the exponentially large Hilbert space to the subspace containing only a single excitation. The corresponding Hamiltonian then reads

$$H = \sum_{i=1}^{N} E_i\, |i\rangle\langle i| + \sum_{i\neq j}^{N} V_{i,j}\, |i\rangle\langle j|, \tag{2}$$

where $|i\rangle$ denotes the state where the excitation is located at site 'i'. In the following, we will assume the energies of all sites to be identical; hence, we may set $E_i = 0$. Concerning the coupling $V_{i,j}$ between site 'i' and 'j', we assume a dipole–dipole interaction of the following form:

$$V_{i,j} = c\, r_{i,j}^{-3} \tag{3}$$

with constant $c \in \mathbb{R}$. For simplicity, we neglect any angular dependences, because, although the excitonic energy transport properties of a particular configuration are affected by these, the influence on the statistics of transport is, as we have checked, rather weak. Thus, the coupling $V_{i,j}$ solely depends on the distance $r_{i,j} = |\boldsymbol{r}_i - \boldsymbol{r}_j|$ between the positions $\boldsymbol{r}_i$ and $\boldsymbol{r}_j$ of the sites i and j.

We identify the initially populated input site 'in' with the first site '1'. The sites '2' to '$N-1$' are the 'intermediates' and 'N' is the designated output site, 'out'. Central to our studies is the efficiency of transport from 'in' to 'out' which, in turn, is governed by the positions of the intermediate sites. For the following statistical analysis, we will consider a random distribution of these positions, where the input and output sites are located at the poles of a sphere, whereas the intermediates are distributed uniformly in the sphere's interior. Note that the Hamiltonian and, in particular, the efficiency of a configuration are invariant with respect to the rotations of the intermediate sites around the polar axis.

3. Transport efficiency

As a quantitative measure for the transfer efficiency, we have considered, in a previous paper (Scholak *et al* 2011), the maximal output probability $\mathcal{P}$ during a short time window $\mathcal{T}$; see section 3.1 below. This measure quantifies transfer within the molecular network from the input to the output site. The transfer is regarded as successful if the excitation reaches the output site with high probability in a short time. A second, complementary measure, which we will introduce in section 3.2, additionally takes into account the subsequent processing of the excitation once it has reached the output site—which may take a significantly longer time than just the transfer from 'in' to 'out'. To provide a complete description of the transfer process, we will, in the following, discuss both these measures and compare them with each other.

3.1. Short-time efficiency $\mathcal{P}$

As a speed benchmark for the excitation's journey from the input to the output site, we consider the time the excitation transfer would take if only those two sites were present. In

this case, the excitation performs a clean, perpetual Rabi oscillation with amplitude 1 and frequency $|V_{\text{in,out}}|$. Hence, a measurement at

$$T = \frac{\pi}{2|V_{\text{in,out}}|} \tag{4}$$

detects the excitation at the destination with probability 1.

Our objective is to beat this time by taking advantage of the additional intermediate sites: due to their smaller distances, these will be coupled more strongly than 'in' and 'out'. To quantify the efficiency of a particular setup of sites, we define a time interval $[0, \mathcal{T}]$, during which we monitor the population of site 'out', p_{out}, and record its maximal value:

$$\mathcal{P} = \max_{t\in[0,\mathcal{T}]} p_{\text{out}}(t). \tag{5}$$

Henceforth, the time

$$t_{\mathcal{P}} = \min\{t \in [0, \mathcal{T}] | p_{\text{out}}(t) = \mathcal{P}\} \tag{6}$$

denotes the smallest time at which the output probability $\mathcal{P}$ is reached. In a closed system, $p_{\text{out}}(t)$ is oscillating indefinitely and a small value of $\mathcal{P}$ does not prevent p_{out} from growing beyond $\mathcal{P}$ at times $t > \mathcal{T}$. Below, we will choose $\mathcal{T}$ significantly smaller than T of equation (4) but large enough compared to the smallest time scale of the system dynamics, to permit complete transfer of the excitation.

3.2. Transfer time $\mathfrak{T}$

The transfer to a so-called 'sink' is another possible efficiency benchmark. The sink is by definition a reservoir that is brought into contact with the system at the site 'out' and may absorb energy from the latter. Any energy that enters the sink will never re-enter the molecular network. Since this—on a phenomenological level—mimics the delivery of the excitation to the reaction centre, many authors (Beatty *et al* 2005, Olaya-Castro *et al* 2008, Rebentrost *et al* 2009) have introduced the sink population to judge the transfer efficiency.

Let Γ be the rate of the trapping process. The Markovian master equation describing the drain to the sink reads (Kenkre *et al* 1982)

$$\begin{aligned}\dot{\varrho}(t) = &-\mathrm{i}\,[H, \varrho(t)] + \Gamma\big(|0\rangle\langle\text{out}|\varrho(t)|\text{out}\rangle\langle 0| \\ &- \tfrac{1}{2}\,|\text{out}\rangle\langle\text{out}|\varrho(t) - \tfrac{1}{2}\,\varrho(t)|\text{out}\rangle\langle\text{out}|\big),\end{aligned} \tag{7}$$

where $\langle\text{out}|\varrho(t)|\text{out}\rangle = p_{\text{out}}(t)$ is the output population and $|0\rangle$ is the ground state to which the system relaxes due to the coupling to the sink. The non-unitary part of equation (7) will be referred to as $L_{\text{sink}}(\varrho(t))$, i.e. the Lindbladian superoperator of steady sink drain.

The asymptotic sink population,

$$\lim_{t\to\infty} \mathfrak{p}(t) = \lim_{t\to\infty} \langle 0|\varrho(t)|0\rangle, \tag{8}$$

is a typical long-time efficiency measure but proves to be useless for our statistical analysis. As we find, for almost all configurations, $\mathfrak{p}(t)$ converges to 1 as the time t tends to infinity. In principle, there exist also configurations exhibiting a so-called 'orthogonal subspace' that prevents the excitation from ever filling the sink—this appears under the name 'coherent population trapping' in the literature, see for instance (Arimondo and Orriols 1976)—but realizations of disorder rigorously featuring this characteristic occur with probability zero.

Under this circumstance, a meaningful measure is the average transfer time,

$$\mathfrak{T} = \int_0^\infty t\,\dot{\mathfrak{p}}(t)\,\mathrm{d}t = \Gamma\int_0^\infty t\,p_{\text{out}}(t)\,\mathrm{d}t, \tag{9}$$

where we used $\dot{\mathfrak{p}}(t) = \langle 0|\dot{\varrho}(t)|0\rangle = \Gamma p_{\text{out}}(t)$, according to equation (7). In the simple case where the system is just composed of the in- and output sites, the transfer time amounts to

$$\mathfrak{T} = \frac{2}{\Gamma} + \frac{\Gamma\,T^2}{\pi^2}. \tag{10}$$

The physics covered by this expression can be extrapolated to larger systems. If Γ^{-1} is very long compared to any system-specific time scale (weak trapping regime), the sink will be an invisible spectator that monitors the system dynamics without significant perturbation. Transport to the sink will be very slow, but steady. If, in contrast, Γ is very large (strong trapping regime), dynamics involving the site 'out' will be slowed down or even halted due to the quantum Zeno effect (Misra and Sudarshan 1977)—a term coined in reference to ancient Greek philosopher Zeno of Elea and his famous paradoxes. In between, there is an optimal trapping rate, as shown below.

4. Optimal configurations

The transfer efficiencies defined above sensitively depend on the particular Hamiltonian of the network, i.e. on the spatial arrangement of the sites and the form of the potential $V_{i,j}$. Of special interest are the optimal configurations which maximize the output probability $\mathcal{P}$ and those which minimize the transfer time $\mathfrak{T}$. In order to find these optimal configurations, we employ an already existing implementation of a global, non-deterministic numerical optimization method (Auger and Hansen 2005), which is based on the evolutionary mechanisms: mutation, selection and adaptation. Note that the configurations presented here are optimized using the isotropic interaction defined in equation (3) and will not necessarily retain their optimal efficiency if the angular dependence of the dipole–dipole interaction is included.

4.1. Maximum short-time efficiency $\mathcal{P}$

In figure 1, we show the maximal transfer efficiencies $\mathcal{P}$ as a function of the monitoring time $\mathcal{T}$, for different numbers N of sites. In other words, the genetic algorithm here finds those configurations which mediate the highest yield at a time $t_{\mathcal{P}} \leqslant \mathcal{T}$. Every point in this plot is the result of many different runs of the optimization routine. Since the algorithm might get trapped in a local minimum, the achieved best values for $\mathcal{P}$ are, strictly speaking, only lower bounds, which we expect to be less tight for larger N. First, we note that perfect transfer, $\mathcal{P} = 100\%$, can indeed be achieved—here actually up to numerical accuracy and provided the monitoring time is not too short. How short this time may be depends, in turn, on the number of sites: as evident from figure 1, complete

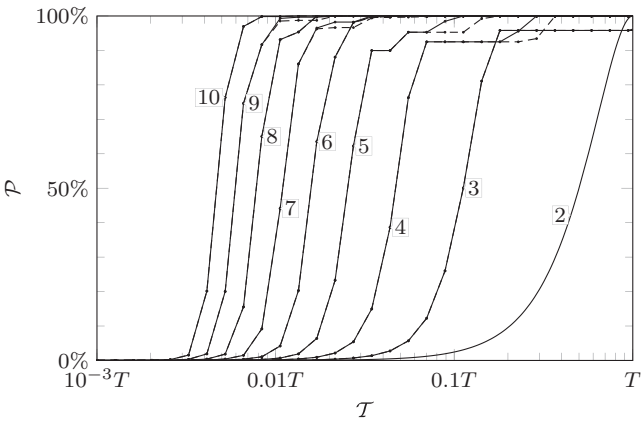

Figure 1. Maximum transfer efficiency $\mathcal{P}$ as a function of the monitoring time $\mathcal{T}$ for $N = 2, \ldots, 10$ sites. Perfect transfer, $\mathcal{P} = 100\%$, can be realized at ever shorter times with increasing N. The dashed lines show the maximum transfer efficiency restricted to one-dimensional spatial molecular configurations, which, for a wide range of parameters, almost reach the optimum realized by three-dimensional configurations (solid lines).

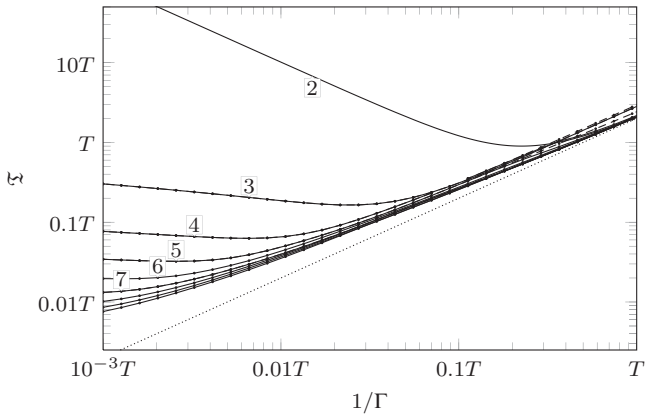

Figure 2. Minimum transfer time $\mathfrak{T}$ as a function of the inverse sink rate Γ for $N = 2, \ldots, 10$ sites. For small Γ (i.e. large $1/\Gamma$), $\mathfrak{T}$ is close to $2/\Gamma$ (dotted line) and approximately independent of N. With increasing N, this behavior holds for increasingly larger values of Γ, such that shorter transfer times can be reached. Similarly as in figure 1, the dashed lines show the minimum transfer time restricted to 1D configurations, which only very slightly deviates from the 3D optimum.

transfer, $\mathcal{P} = 100\%$, for $N = 5$ sites is achieved ten times faster compared to the Rabi period T, equation (4), for $N = 2$ sites, and hundred times faster for $N = 9$ sites. This clearly shows that optimal transport can be accelerated by the addition of sites. As the plateaus visible for $N = 3$ and $N = 4$ (and, though less pronounced, also for larger N) indicate, even faster transfer times are possible if one admits small deviations, e.g. $\mathcal{P} > 90\%$, from perfect transport.

As mentioned above, the optimal efficiency $\mathcal{P}$ is determined by finding the optimal molecular configuration in three-dimensional space. For comparison, the dashed lines in figure 1 show the optimal transfer efficiency restricted to one-dimensional configurations, where all sites are located on the straight line between the input and the output site. As we see, for a wide range of parameters, the optimum can almost be reached in 1D. A closer inspection reveals only tiny differences between the solid and dashed lines of the order of 10^{-3}. This shows that, typically, the optimal configuration is close to—though not exactly—one dimensional, as in the example shown in figure 3(a), which is neither 1D nor exactly 2D and realizes perfect transport, $\mathcal{P} = 100\%$, at the shortest time $t_{\mathcal{P}}^{(\mathrm{opt})} = 0.0339T$ for $N = 7$ sites. Intuitively, this is explained by the fact that the mutual distances between the sites are smallest for one-dimensional configurations, thus enabling larger coupling and faster transport (Ates 2009). Let us point out, however, that in some cases, for example for $N = 4$, 5 and $\mathcal{P} \simeq 0.95$ in figure 3(a), larger deviations from 1D are required to reach the optimum. Furthermore, if we relax the constraint that the transfer time for $\mathcal{P} = 100\%$ be minimal and allow, e.g., $\mathcal{T} = 0.1T$ for $N = 7$, a large variety of configurations providing complete transport from 'in' to 'out' are found, most of which are manifestly 3D and very distinct from lattice-like or collinear structures; see the example shown in Scholak *et al* (2011).

4.2. *Minimum transfer time* $\mathfrak{T}$

The analogous analysis for $\mathfrak{T}$ is shown in figure 2, where the minimum transfer time is plotted as a function of the sink rate Γ, for different numbers N of sites. As already discussed in section 3.2, for small Γ, the transfer time is limited mainly by the coupling to the sink, and approximately proportional to $1/\Gamma$ (dotted line). In this regime, the transfer can hence be accelerated by increasing Γ up to its optimum value, which is roughly given by the internal timescale of transfer from 'in' to 'out'. For $N = 2$, according to equation (10), this is the case for $\Gamma = \sqrt{2}\pi/T$, leading to $\mathfrak{T} = 2\sqrt{2}T/\pi$. For larger N, as the internal transfer from 'in' to 'out' for optimal configurations becomes faster, the minimum is shifted to larger values of Γ, with correspondingly shorter transfer times.

Again, the dashed lines in figure 2 display the minimum transfer time restricted to 1D configurations. Here, hardly any difference from the 3D case is visible. This indicates that the configurations optimized with respect to $\mathfrak{T}$ are even closer to 1D than those optimized with respect to $\mathcal{P}$. As an example, figure 3(d) displays the optimal spatial configuration for $N = 7$ and $\Gamma = 10/T$, which achieves the minimum transfer time $\mathfrak{T} = 0.266T$.

For the relatively moderate value of the sink rate $\Gamma = 10/T$, a comparison between the fully coherent dynamics and the dynamics in the presence of the sink, for the two configurations optimized with respect to $\mathcal{P}$ on the one hand and with respect to $\mathfrak{T}$ on the other hand, is shown in figure 3. Let us first look at the coherent dynamics; see figures 3(b) and (e). The configuration in 3(a) has been optimized with respect to $\mathcal{P}$ and achieves $\mathcal{P} = 100\%$ already at $t_{\mathcal{P}} = 0.339\mathcal{T}$, where the time window $\mathcal{T}$ is chosen as $\mathcal{T} = \Gamma^{-1} = T/10$. On the other hand, the configuration in (d), that has been optimized with respect to the transfer time $\mathfrak{T}$, achieves only $\mathcal{P} = 60.1\%$ during the time window $\mathcal{T}$—although $p_{\mathrm{out}}(t)$ (solid line in figure 3(e)) increases up to a maximum value of

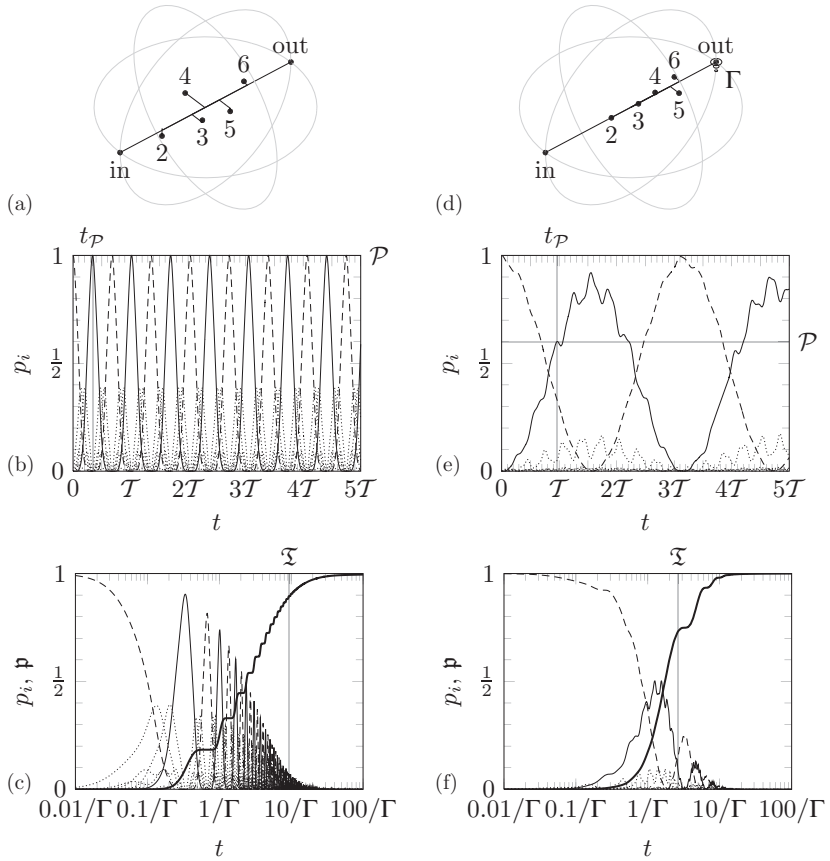

Figure 3. Spatial configurations as well as population and sink filling dynamics for two exemplary configurations, (a) and (d), with $N = 7$ sites which are optimized with respect to the transfer efficiency $\mathcal{P}$ (time window $\mathcal{T} = T/10$), on the one hand (left column), and with respect to the transfer time $\mathfrak{T}$ (sink rate $\Gamma = \mathcal{T}^{-1} = 10/T$), on the other hand (right column). The arrangement shown in (a) is also optimal with respect to $t_{\mathcal{P}}$: it achieves $\mathcal{P} = 100\%$ at the shortest time $t_{\mathcal{P}}^{(\mathrm{opt})} = 0.339\mathcal{T} = 0.0339T$. Whereas the intermediate sites in configuration (a) are slightly out of plane, the configuration (d) is strictly two dimensional. The middle row (b, e) shows the populations of site 'out' (solid line), site 'in' (dashed line) and the intermediate sites (dotted lines) for purely coherent dynamics, whereas the lower row (c, f) additionally displays the filling $\mathfrak{p}(t)$ (thick solid line) in the case of sink dynamics. For a drain rate of $\Gamma = 10/T$, the configuration (d) has an optimally short transfer time of $\mathfrak{T} = 2.66/\Gamma$. However, (d) is not optimal with respect to $\mathcal{P}$, as it achieves only $\mathcal{P} = 60.1\%$ in the monitored time window. In turn, the configuration (a) is less efficient than (d) in terms of sink transfer, since it exhibits a transfer time of $\mathfrak{T} = 9.42/\Gamma$, see (c).

about 90% at a later time $t < 2\mathcal{T}$, where it exhibits a broad maximum. Remarkably, $p_{\mathrm{in}}(t)$ (dashed line) returns to its initial value 1 at $t \simeq 3\mathcal{T}$. This renders the coherent dynamics close to periodic, such that a broad maximum of $p_{\mathrm{out}}(t)$ occurs repeatedly. In the corresponding sink dynamics, see figure 3(f), we see that the sink is almost completely filled after the third oscillation, and already three-quarter filled after the first oscillation. Thereby, the minimum transfer time $\mathfrak{T} = 2.66/\Gamma$ is achieved. In contrast, for configuration (a), although the oscillations are faster and the filling of the sink starts much earlier, the filling is less efficient, see figure 3(c), leading to a longer transfer time of $\mathfrak{T} = 9.42/\Gamma$.

Another difference between the optimal configurations (a) and (d) is as follows: whereas for configuration (a), all the intermediate sites 2–6 are successively populated during the transfer from 'in' to 'out', basically only four sites ('in', 2, 4 and 'out') participate in the case of configuration (d). Accordingly, figure 2 shows that the minimum transfer time at $\Gamma = 10/T$ is almost the same for $N = 4$ as for $N = 7$. This, however, changes for larger sink rates Γ. For example, in the case $N = 7$—on which we will focus in the following, since transport in the FMO complex occurs between seven different molecules—the optimal sink rate is $\Gamma \simeq 10^3/T$, with transfer time $\mathfrak{T}^{(\mathrm{opt})} \simeq 0.0132T$. Here, we find the corresponding optimal configuration to be perfectly one-dimensional, with almost equal distances between the sites. This gives rise to a fast, ballistic transport from one site to the other, until the excitation reaches 'out' where it is almost immediately absorbed. Again, at this large value of Γ, the transfer time of configuration (a) is significantly longer, i.e. $\mathfrak{T} = 0.264T \simeq 20\mathfrak{T}^{(\mathrm{opt})}$.

4.3. Stability

Spatial robustness is a criterion that we do not take into account for the optimization, even though—from the point of view of natural evolution—stability is certainly important. Convergence of the optimization algorithm will only be possible if the efficiency landscape is sufficiently smooth, i.e. if in the close vicinity of an exceptionally efficient configuration,

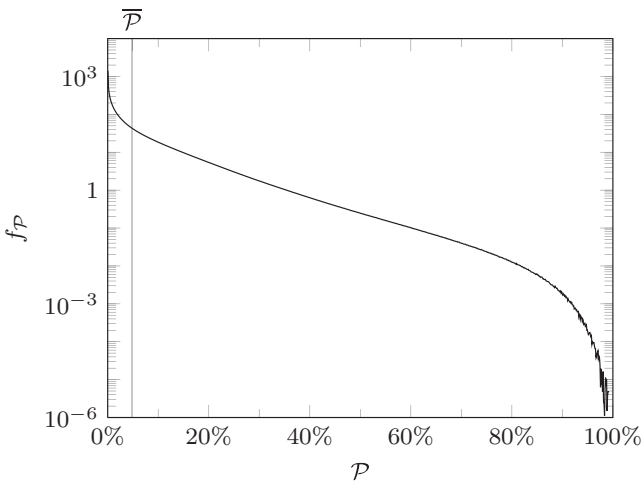

Figure 4. Statistics of the transfer efficiency $\mathcal{P}$ for $N = 7$ and $\mathcal{T} = T/10$. Shown is a logarithmic plot of the probability density $f_{\mathcal{P}}$. The expectation value is $\overline{\mathcal{P}} \simeq 4.85\%$, and configurations with high efficiency are rare.

there is no completely inefficient one. Therefore, we know that the obtained optimal configurations are, in principle, stable with respect to static spatial noise.

The stability of a particular optimal configuration can be assessed by introducing spatial mutations in a controlled fashion. For this purpose, we subject each intermediate site to small positional fluctuations, Gaussian distributed with standard deviation σ (normalized to the radius of the sphere, $r_{\text{in,out}}/2$), and record the mean values $\langle\mathcal{P}\rangle$ and $\langle\mathfrak{T}\rangle$ under these static fluctuations. For optimal configurations, we find a quadratic dependence for small σ. For example, for the two configurations shown in figure 3: $1-\langle\mathcal{P}\rangle = (34\sigma)^2$ for sample (a) and $\langle\mathfrak{T}\rangle/[\mathfrak{T}]_{\sigma=0} - 1 = (65\sigma)^2$ for configuration (d).

5. Statistical correlations

As seen above, a configuration optimized with respect to the transfer efficiency $\mathcal{P}$ is not necessarily optimal with respect to the transfer time $\mathfrak{T}$, and vice versa. Nevertheless, we expect that these two transport measures are in some way correlated with each other. A statistical analysis of these correlations will be performed in this section. The statistical ensemble used for this purpose is generated by the uniform distribution of sites, as described in section 2. As a time window for the transfer efficiency $\mathcal{P}$, we choose, as in figure 3, $\mathcal{T} = T/10$, and, correspondingly, $\Gamma = \mathcal{T}^{-1} = 10/T$ for the sink rate. The number of sites is set to $N = 7$.

First, figure 4 shows the probability density $f_{\mathcal{P}}$ of $\mathcal{P}$. As already discussed in Scholak *et al* (2011), the vast majority of configurations are inefficient, such that the mean value $\overline{\mathcal{P}} = \int_0^1 x\, f_{\mathcal{P}}(x)\,\mathrm{d}x \simeq 4.85\%$ is quite low. However, a small fraction of configurations—one out of 2×10^5—exhibits $\mathcal{P} > 90\%$.

In figure 5(a), we display the probability density $f_{\mathfrak{T}}$ of $\mathfrak{T}$. The curve begins abruptly at the minimum transfer time $\mathfrak{T} = 2.66/\Gamma$ (see section 4.2). After a steep ascent, the probability density distribution peaks at its most probable value, $\mathfrak{T} \simeq 10/\Gamma = T$. The subsequent descent is algebraic in $\mathfrak{T}$, i.e. as $x \to \infty$, $f_{\mathfrak{T}}(x) \propto x^{-\alpha}$ with $\alpha \simeq 1.4$. Therefore, neither the mean value nor the variance of $\mathfrak{T}$ exists, or, in other words, extremely long transfer times occur with non-negligible probability. Therefore, we will use the median $\tilde{\mathfrak{T}}$, defined by $\int_0^{\tilde{\mathfrak{T}}/T} f_{\mathfrak{T}}(x)\,\mathrm{d}x = 1/2$, in order to define a typical value for the transfer time, and find: $\tilde{\mathfrak{T}} \simeq 2920/\Gamma$.

As already mentioned in section 3.2, long transfer times are caused by the eigenstates $|\sigma\rangle$ of the molecular Hamiltonian, which are almost orthogonal to the site 'out' ($\langle\text{out}|\sigma\rangle \simeq 0$) but have a finite overlap with the initial state $|\text{in}\rangle$ ($\langle\text{in}|\sigma\rangle > 0$). The space spanned by these states is an orthogonal subspace $\mathcal{O}_{\text{out}}$ with respect to $|\text{out}\rangle$. $\mathcal{O}_{\text{out}}$ is contained in the kernel of the sink's Lindbladian super-operator, L_{sink}, i.e. any state inside $\mathcal{O}_{\text{out}}$ will never enter the sink. Note that—in our statistical ensemble—the probability of occurrence of a strict orthogonal subspace is zero. The residual overlap of states $|\sigma\rangle \in \mathcal{O}_{\text{out}}$ and $|\text{out}\rangle$ may become arbitrarily small, though, and therefore lead to arbitrarily long transfer times. We will use the term 'orthogonal subspace' in this weak sense.

In order to unravel the correlations between $\mathcal{P}$ and $\mathfrak{T}$, figure 5(b) shows a density plot of the transfer time distribution $f_{\mathfrak{T}|\mathcal{P}}$ which is conditioned on values of $\mathcal{P}$. Clearly, there is an overall tendency to favour small $\mathfrak{T}$ if $\mathcal{P}$ is large, e.g. the median $\tilde{\mathfrak{T}}$ of $\mathfrak{T}$ at given $\mathcal{P}$ (white line) is, on average, a decreasing function of $\mathcal{P}$. This tendency, however, is not strict: for example, there are also configurations that achieve high output probability in $[0, \mathcal{T}]$ but nevertheless exhibit large transfer times. This is brought about by the orthogonal subspaces that, in general, affect the transfer time much more drastically than the maximal output probability. Even a tiny population of blocking eigenstates causes $\mathfrak{T}$ to explode, whereas, for the same configuration, $\mathcal{P}$ still can be close to 1. Conversely, small $\mathcal{P}$ does not necessarily imply long $\mathfrak{T}$. This is due to the definition of $\mathcal{P}$, see equation (5), according to which only times smaller than $\mathcal{T}$ are considered. Therefore, $\mathcal{P}$ does not account for those configurations where probability builds up at 'out' at somewhat later times in order to increase the sink drain.

In figure 5(c) we see the probability density $f_{\mathcal{P}|\mathfrak{T}}$ and the mean value $\overline{\mathcal{P}}$ both of which are conditioned on values of $\mathfrak{T}$. Here, the conclusions are similar as for figure 5(b): although there is certainly not a one-to-one correspondence between $\mathcal{P}$ and $\mathfrak{T}$, we see that, on average, $\mathcal{P}$ monotonically increases as $\mathfrak{T}$ decreases. Indeed, small values of $\mathfrak{T}$ imply larger, though not saturated, maximal output probabilities $\mathcal{P}$. The latter point agrees with section 4, where we noted that configurations that exhibit optimal transport with respect to $\mathfrak{T}$ will not reach 100% output population, if the excitation is propagated without drain. Furthermore, it is also clearly visible that very small $\mathcal{P}$ are impossible, if $\mathfrak{T}$ is minimal.

6. Efficiency and dephasing noise

So far, we have discussed excitation transfer solely for fully coherent dynamics or in the presence of a sink linked to the site 'out'. In reality, however, the internal electronic levels of the FMO complex are coupled to a multitude of uncontrolled

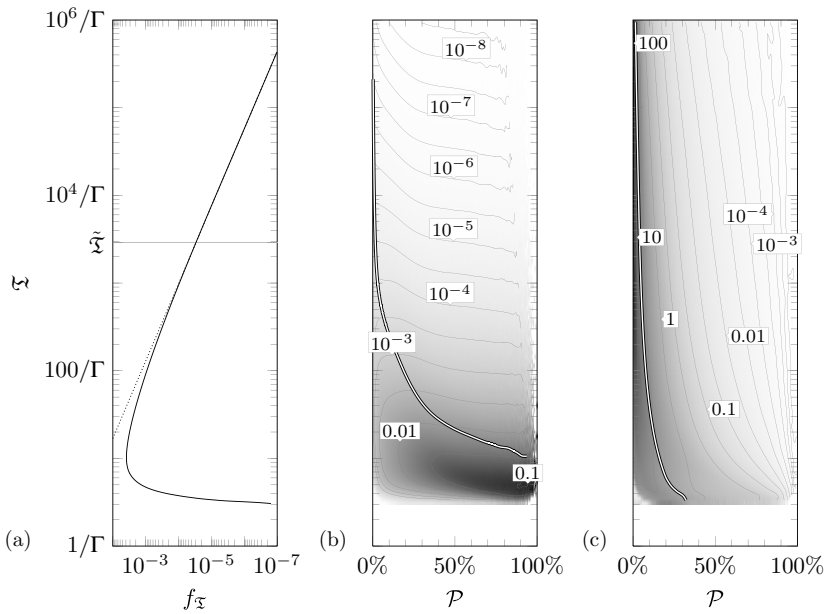

Figure 5. Statistics of the average sink transfer time $\mathfrak{T}$ for $N = 7$ and $\Gamma = 10/T$. (a) Probability density $f_{\mathfrak{T}}$ with its subexponential decay, $f_{\mathfrak{T}}(x) \propto x^{-1.38}$ (dotted line). The median $\tilde{\mathfrak{T}}$ is approximately $2920/\Gamma$. (b) Conditional probability density $f_{\mathfrak{T}|\mathcal{P}}$ as a function of $\mathcal{P}$. The median of $\mathfrak{T}$ (white line) decreases as $\mathcal{P}$ increases. (c) Conditional probability density $f_{\mathcal{P}|\mathfrak{T}}$ and average transfer efficiency $\overline{\mathcal{P}}$ (white line) as a function of $\mathfrak{T}$. Again, $\overline{\mathcal{P}}$ increases with decreasing $\mathfrak{T}$. This shows that, on average, smaller transfer times $\mathfrak{T}$ imply larger efficiencies $\mathcal{P}$ and vice versa.

external degrees of freedom. Therefore, in this section, we will extend our statistical analysis in order to account for the possible impact of environmental noise. We summarize the alien degrees of freedom as an abstract object, as 'the environment', that affects all sites and deteriorates quantum coherence. In particular, we will clarify under what conditions the excitation transfer is favoured or hindered by the presence of dephasing noise.

For this purpose, we will use, as the simplest, paradigmatic model for dephasing, the following Lindbladian super-operator (Breuer and Petruccione 2002):

$$L_{\text{deph}}(\varrho) = -4\gamma \sum_{i \neq j}^{N} |i\rangle\langle i|\varrho|j\rangle\langle j|, \tag{11}$$

which describes global and homogeneous dephasing between different sites with rate 4γ. The dynamics of the reduced density matrix ϱ of the electronic degrees of freedom is then determined by the Markovian master equation $\dot{\varrho}(t) = -\mathrm{i}\,[H, \varrho(t)] + L(\varrho(t))$, where $L = L_{\text{deph}}$ or $L = L_{\text{sink}} + L_{\text{deph}}$, cf equation (7), for the transfer efficiency $\mathcal{P}$ or the transfer time $\mathfrak{T}$, respectively.

As well known for a pure dephasing model like equation (11), in the limit $\gamma \to \infty$ all dynamics is eventually halted due to the quantum Zeno effect (Misra and Sudarshan 1977). More precisely, for large, but finite dephasing rates, the evolution of the populations $\varrho_{k,k}$ is approximated by the solutions of a classical rate equation,

$$\dot{\varrho}_{k,k}(t) = \sum_{l} [W_{k,l}\, \varrho_{l,l}(t) - W_{l,k}\, \varrho_{k,k}(t)], \tag{12}$$

where the matrix W is assembled from the transition rates $W_{k,l} = |V_{k,l}|^2/(2\gamma)$. Note that the dephasing rate is in the denominator of the $W_{k,l}$. Hence, the larger its value, the slower the system dynamics—a precursor of the Zeno freezing effect.

6.1. Statistics of the short-time efficiency $\mathcal{P}$

Figure 6 shows a contour plot of the probability density $f_{\mathcal{P}}$ as a function of the dephasing rate γ. We sample the whole landscape, from the regime of almost coherent evolution ($\gamma T \ll 1$) to the 'Zeno valley' ($\gamma T \gg 1$), where quantum evolution is strongly impeded by the quantum Zeno effect. In addition, the figure contains the average efficiency $\overline{\mathcal{P}}(\gamma)$ (white line), the efficiency of the optimal configuration of figure 3(a) (dashed line), and the maximum efficiency (dot-dashed line) as a function of γ. The latter is obtained by numerical optimization of the spatial configuration with respect to $\mathcal{P}$, cf section 4, at 200 different values of the dephasing rate γ.

For γ below $0.01/\mathcal{T}$, the probability density $f_{\mathcal{P}}$ is almost unchanged with respect to its fully coherent counterpart; see figure 4. With increasing γ, however, the phase noise more and more corrupts the short-time transport efficiency, $\mathcal{P}$. As evident from figure 6, especially large efficiencies become more and more improbable at larger γ. In particular, the efficiency of the configuration shown in figure 3(a) decreases monotonically as a function of γ (dashed line). Interestingly, this configuration—the optimal one for $\gamma = 0$—remains among the most efficient configurations for a large range of γ's, although for $\gamma > 0.1$, the maximum efficiency (dot-dashed line) is reached by other configurations, which are even more robust with respect to dephasing. On the other hand, the probability density at extremely low efficiencies, i.e. $\mathcal{P} < 0.1\%$, also drops for small γ. As already pointed out in our previous work (Scholak *et al* 2011), this implies that

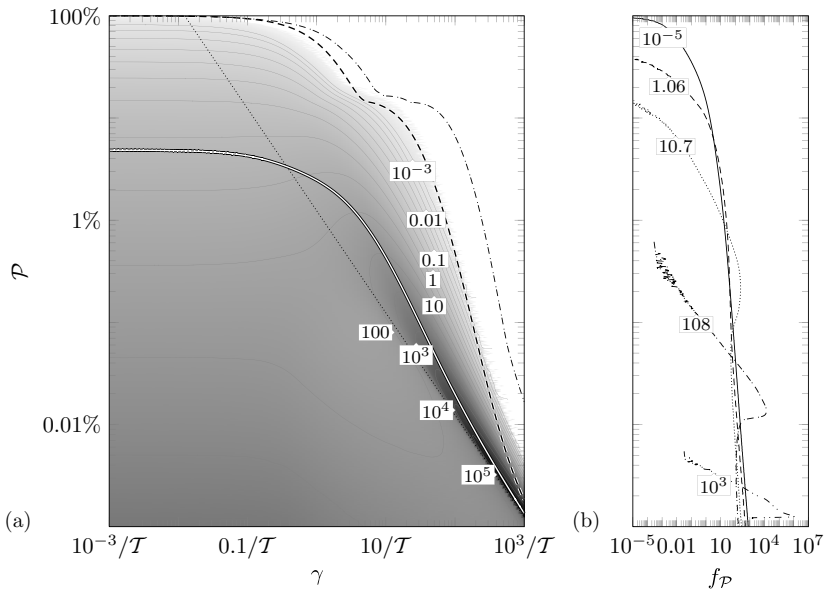

Figure 6. (a) Probability density $f_{\mathcal{P}}$, for $N = 7$ and $\mathcal{T} = T/10$, as a function of the dephasing rate γ. In total, the efficiency decreases with increasing γ. In particular, the average efficiency $\overline{\mathcal{P}}$ (white line), the maximal efficiency $\mathcal{P}$ (dot-dashed line), and the efficiency of the configuration shown in figure 3(a) (dashed black line) are all monotonically decreasing functions of γ. For large γ, the efficiency decays inversely proportional to γ (dotted line). (b) Some vertical cross sections of the landscape. The probability density $f_{\mathcal{P}}$ is displayed for $\gamma = 10^{-5}/\mathcal{T}$ (solid), $1.06/\mathcal{T}$ (dashed), $10.7/\mathcal{T}$ (dotted), $108/\mathcal{T}$ (dot-dashed), and $10^{3}/\mathcal{T}$ (dot-dot-dashed).

configurations with low efficiency may also benefit from the presence of noise.

At very large γ, the Zeno effect sets in (see above)—here, the average efficiency is inversely proportional to γ (dotted line), and the probability distribution gets more and more peaked around this value. This decrease is symptomatic for overdamped dynamics, where the system approaches the stationary state with transition rates inversely proportional to γ, cf equation (12).

In total, we stress that—in spite of some low-efficiency configurations which profit from dephasing—the short-time efficiency $\mathcal{P}$ on average decreases with increasing dephasing (white line in figure 6). This can be understood from the fact that, in the case of coherent transport, the output population is subject to quantum interference between many path amplitudes leading from 'in' to 'out'. This interference may be either constructive or destructive and, accordingly, the suppression of interference by dephasing may either diminish or enhance transport. Since the efficiency $\mathcal{P}$ selects the point of maximally constructive interference in the time window $[0, \mathcal{T}]$ (see the definition in equation (5)), $\mathcal{P}$ is likely to be degraded by dephasing—especially, if the efficiency without dephasing is large.

6.2. *Statistics of the transfer time* $\mathfrak{T}$

As shown below, the situation will be different for the transfer time $\mathfrak{T}$, which—in contrast to $\mathcal{P}$—is sensitive to dynamics on time scales longer than $\mathcal{T}$.

Figure 7 shows the probability density of $\mathfrak{T}$, for $N = 7$ and sink rate $\Gamma = 10/T$, as a function of the dephasing rate γ. As we have seen in figure 5(a), for $\gamma = 0$, the probability density of the transfer time exhibits an abrupt increase at the minimum transfer time $2.66/\Gamma$, has its most probable value at $\mathfrak{T} \simeq 10/\Gamma$, and then decays subexponentially, i.e. has a heavy tail. For weak dephasing, the weight of this tail is gradually decreased, and thus very long transfer times are less probable than without dephasing. In other words, the differences between spatial arrangements become less and less important. This is due to the fact that dephasing makes it possible to escape from the orthogonal subspaces, which, as noted in section 5, are responsible for very long transfer times.

This process is accompanied by a slight clustering of transfer times around a particular value that depends inverse linearly on the dephasing rate, as indicated by the downward dotted line on the left side of the plot. In figure 7(b), this phenomenon is visible as a cone-shaped, narrow peak at $\mathfrak{T} = 1/(4\gamma)$. As we have found, the configurations that give rise to transfer times in the vicinity of this peak feature one eigenstate in the orthogonal subspace $\mathcal{O}_{\text{out}}$, which is mainly localized on two sites. This happens, typically, for configurations where one site is in close (but not immediate) neighbourhood of the site 'in', and all other sites are approximately equally distant from these two, i.e. either far away or in the centre plane defined by these two sites. On the other hand, transfer times longer than those of the above peak are typically caused by configurations with one site placed in the immediate neighbourhood of site 'in' (or 'out'), thus giving rise to two (or $N - 2$) blocking eigenstates, respectively.

On the other side of the picture, for $\gamma > 10\Gamma$, the transfer time increases linearly with γ. As already discussed, this is a consequence of the quantum Zeno effect or, more precisely, of the fact that, for large γ, the transport rates decrease inversely proportional to γ, see equation (12).

In between, at $\gamma \simeq \Gamma$, i.e. in the region of highest probability density in figure 7(a), there is a regime of optimal

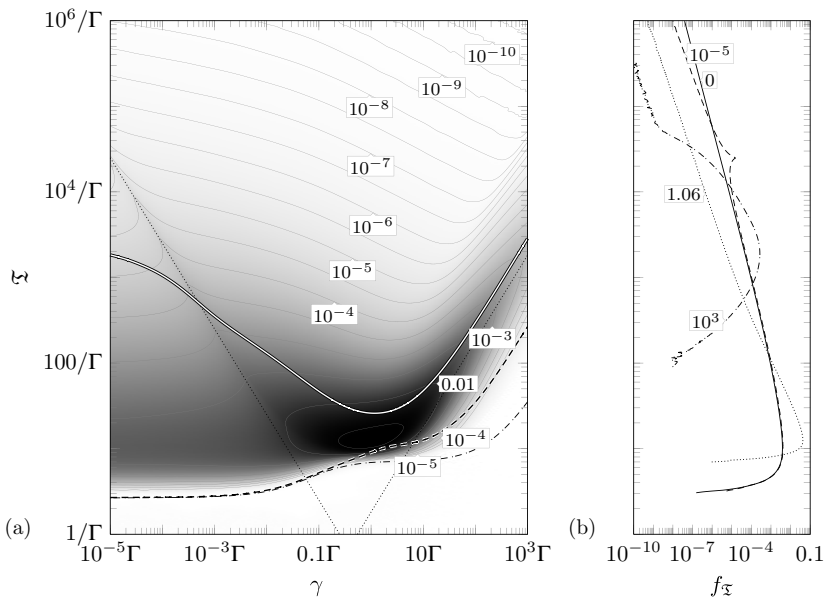

Figure 7. (a) Probability density $f_{\mathfrak{T}}$ for $N = 7$ and sink rate $\Gamma = 10/T$, as a function of the dephasing rate γ. The two dotted, diagonal lines are given by $\mathfrak{T} = (4\gamma)^{-1}$ and $\mathfrak{T} \simeq 1.82\gamma\Gamma^{-2}$, respectively. Furthermore, the white line shows the median $\tilde{\mathfrak{T}}$, the dot-dashed line the minimum transfer time and the dashed line the transfer time of the optimal configuration shown in figure 3(d). Although most configurations exhibit an optimal dephasing rate $\gamma \simeq 1/T$ giving rise to classical transport ($\mathfrak{T} \gg (4\gamma)^{-1}$, i.e. on the right-hand side of the downward dotted line), the globally shortest transfer time is achieved by the optimal configuration at small dephasing rates where transport is coherent (i.e. $\mathfrak{T} \ll (4\gamma)^{-1}$). (b) Vertical cross sections of the landscape in (a): probability densities $f_{\mathfrak{T}}$ for $\gamma = 0$ (solid), $10^{-5}\Gamma$ (dashed), 1.06Γ (dotted), $10^3\Gamma$ (dot-dashed).

dephasing, where the median of the transfer time distribution (white line) assumes its minimum value $\tilde{\mathfrak{T}} \simeq 26/\Gamma$. The probability to randomly sample a configuration, that is as fast as $\mathfrak{T} = 10/\Gamma$, is more than ten times higher when γ is close to Γ than when it is zero. This shows that, for most configurations drawn from the statistical ensemble, adding the right amount of noise helps to accelerate the transfer, in accordance with previous results on environment-assisted transport (Plenio and Huelga 2008, Caruso *et al* 2009, Rebentrost *et al* 2009, Cao and Silbey 2009, Wu *et al* 2010, Chin *et al* 2010). Hence, for a typical configuration, the transfer to the sink is affected by quantum coherence in a destructive rather than constructive way. Extreme examples of destructive interference are given by configurations with orthogonal subspaces, as discussed above. In such (and also other, less extreme) cases, coherence must be broken in order to achieve a faster transfer. Indeed, we see that the dominating black spot in figure 7(a), indicating the region of optimal dephasing, lies on the right-hand side, i.e. classical side of the downward dotted line $\mathfrak{T} \propto \gamma^{-1}$. This line can roughly be interpreted as a border between the regimes of quantum and classical transport: since coherence decays approximately with rate γ, the fact that $\mathfrak{T} \ll \gamma^{-1}$ (or $\mathfrak{T} \gg \gamma^{-1}$) indicates that coherence is still present (or absent) at the time $\mathcal{T}$ when the excitation arrives at the sink.

Let us stress, however, that the above picture does not hold for all configurations, especially not for those which achieve the smallest transfer times. The latter are reached in the lower-left corner of figure 7(a), i.e. in the regime of coherent transport. In particular, the minimum transfer time (dot-dashed line) increases monotonically as a function of γ, as well as the transfer time of the configuration shown in figure 3(d) (dashed line). Interestingly, for dephasing rates smaller than 0.1Γ, the two curves almost coincide (the transfer times deviate by $\simeq 0.01/\Gamma$ for $\gamma = 10^{-5}\Gamma$). This signifies that the configuration of figure 3(d)—albeit optimized for absent noise—remains almost optimal under weak noise. For stronger dephasing, other—although extremely rare—configurations are more efficient. Note that their efficiency, too, is not 'assisted' by the dephasing, since, as we have checked, the transfer times of these configurations as a function of γ are smallest for $\gamma = 0$. They merely stand out due to their robustness against dephasing.

7. Conclusion

As the above results clearly show, the fastest possible transfer of an excitation between two sites of a molecular network is realized without dephasing. Although adding noise accelerates the transfer for most configurations in our statistical ensemble, the globally shortest transfer time is achieved only for special, optimized configurations and purely coherent dynamics. In the absence of noise, these configurations are more than twice as efficient as any environment-assisted configuration giving rise to classical transport even at an optimally chosen dephasing rate.

Whether nature has been able to evolve such optimal configurations and thereby exploit quantum coherence to increase the transfer efficiency, e.g. in the photosynthetic FMO complex, is still an open question. To clarify this, the corresponding molecular Hamiltonian should be characterized as precisely as possible, since, as our statistical studies

indicate, the transport properties sensitively depend on the particular configuration.

And even if the reason for the astonishing efficiency of the FMO complex turns out to be a different one, the theoretical discovery of optimal configurations will surely stimulate efforts to design experiments or technical devices (e.g. organic solar cells) in order to exploit the potentially constructive role of quantum coherence.

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